

#Accenture Coding Class-1 | Questions & Answers

Explanation Video:-

<https://www.youtube.com/live/BZRQhzku7qQ?si=KgWmXn5EwLIHWL0j>

Accenture Previous Year Question Q1

Given an array `arr` of `n` positive integers as its argument. You are Required to calculate the Standard deviation of `N` positive integers in array and return the same.

Standard deviation for data set `arr[0],arr[1],arr[2]...` is equal to the $\sqrt{\frac{[(arr[0]-A)^2+(arr[1]-A)^2+\dots]}{N}}$ where `A` is the average of all The elements of the array. You are required to complete the `StandardDeviation` function which takes `arr` as input and returns ans

Python

```
def StandardDeviation(arr):
```

```
    N=len(arr)
    average=0
    for i in range(N):
        average+=arr[i]
```

```
    average/=N
```

```
    su=0
    for i in arr:
        su+=((i-average)**2)
```

```
    su/=N
    return math.sqrt(su)
```

C++

```
double standardDeviation(const std::vector<int>& arr) {
    int N = arr.size();
    double average = 0;

    // Calculate average
    for (int i = 0; i < N; ++i) {
        average += arr[i];
    }
    average /= N;

    // Calculate sum of squared differences
    double sumOfSquares = 0;
    for (int i = 0; i < N; ++i) {
        sumOfSquares += std::pow(arr[i] - average, 2);
    }

    // Calculate standard deviation
    double variance = sumOfSquares / N;
    return std::sqrt(variance);
}
```

Java

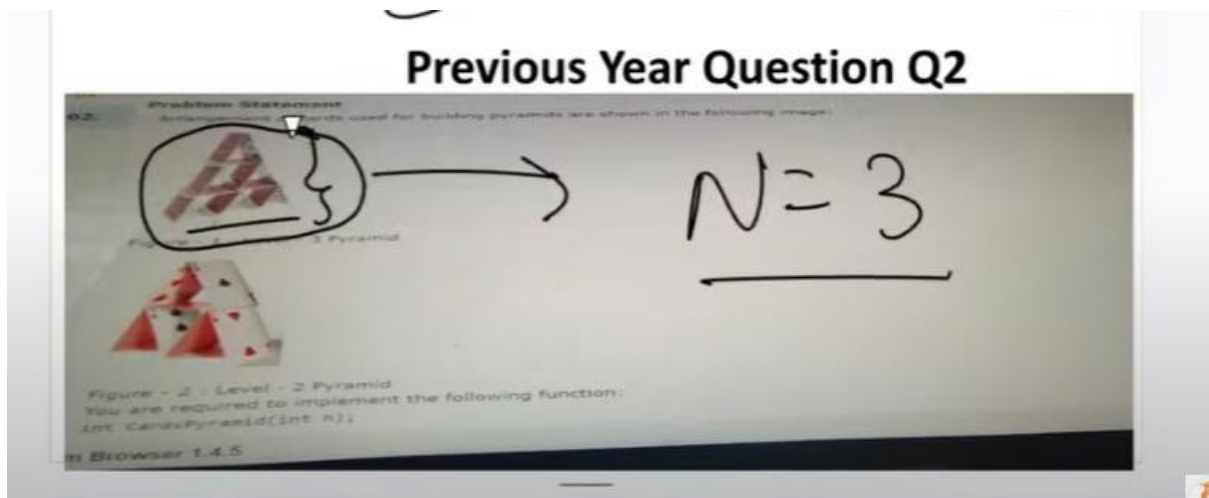
```
public static double standardDeviation(List<Integer> arr) {
    int N = arr.size();
    double average = 0;

    // Calculate average
    for (int i = 0; i < N; ++i) {
        average += arr.get(i);
    }
    average /= N;

    // Calculate sum of squared differences
    double sumOfSquares = 0;
    for (int i = 0; i < N; ++i) {
        sumOfSquares += Math.pow(arr.get(i) - average, 2);
    }

    // Calculate standard deviation
    double variance = sumOfSquares / N;
    return Math.sqrt(variance);
}
```

Q2.



Python

```
def CalculateCards(n):  
    ans=0  
    level=1  
    while(level<=n):  
        ans+=(level*3)  
        level+=1  
    ans-=n  
    return ans
```

```
print(CalculateCards(3))
```

C++

```
int calculateCards(int n) {  
    int ans = 0;  
    int level = 1;  
  
    while (level <= n) {  
        ans += (level * 3);  
        level++;  
    }  
  
    ans -= n;  
    return ans;  
}
```

Java

```
public static int calculateCards(int n) {  
    int ans = 0;  
    int level = 1;  
  
    while (level <= n) {  
        ans += (level * 3);  
        level++;  
    }  
  
    ans -= n;  
    return ans;  
}
```